Continuous Images Preserve Connectedness

An End-to-End Meta-Math Walkthrough

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Abstract

A complete authored walkthrough of continuous images preserve connectedness, with explicit mathematical contracts, local verification gates, and reusable exports. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

GT-01 / General Topology / Report v1

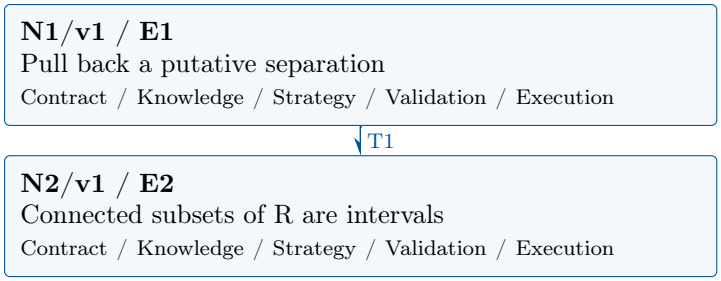
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *For a connected topological space X and continuous $f : X \to Y$, prove $f(X)$ is connected in the subspace topology. Deduce the Intermediate Value Theorem for continuous maps from real intervals to $\mathbb{R}$.*

A. Foundations and permitted imports

A separation of a space is a pair of disjoint nonempty open subsets covering it; connected means there is no separation. Use continuity as openness of inverse images and the subspace topology. Import connectedness of real intervals, as a least-upper-bound consequence; no statement about continuity of an inverse is involved.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1: N1/E1/v1 to N2. Continuous images of connected spaces are connected.

Receiving activity. N2: Connected subsets of R are intervals

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: A connected image in R; prior exports E1/v1

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Pull back a putative separation

Object and role. Work in the actual image, not all of Y.

Inputs. Connected X; continuous $f : X \to Y$; foundational imports stated in §1

C — Contract

Own E1/v1: Continuous images of connected spaces are connected.

K — Knowledge

Retain Connected X; continuous $f : X \to Y$; foundational imports stated in §1; keep their assumptions and versions attached.

S — Strategy

Work in the actual image, not all of Y.

V — Validation

O and P need not be disjoint in Y; their intersections with the image are.

E — Execution

Carry out the numbered deductions and publish E1/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N1.1 Suppose $f(X) = U \cup V$ is a separation in its subspace topology.

N1.2 Write $U = f(X) \cap O$ and $V = f(X) \cap P$ for open $O, P \subset Y$.

N1.3 Their inverse images are open in X by continuity.

N1.4 Both are nonempty because U,V are nonempty subsets of the image.

N1.5 They are disjoint and coverX.

N1.6 This contradicts connectedness.

N1.7 Therefore $f(X)$ is connected; the empty image, if X is empty, also has no separation.

Export E1/v1. Continuous images of connected spaces are connected.

Acceptance gate. O and P need not be disjoint in Y; their intersections with the image are. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N1: O and P need not be disjoint in Y; their intersections with the image are. Recheck dependent nodes N2.

B. N2/v1: Connected subsets of R are intervals

Object and role. Use a missing intermediate point to create a separation.

Inputs. A connected image in R; prior exports E1/v1

C — Contract

Own E2/v1: A topological proof of the intermediate value theorem.

K — Knowledge

Retain A connected image in R; prior exports E1/v1; keep their assumptions and versions attached.

S — Strategy

Use a missing intermediate point to create a separation.

V — Validation

Connectedness of the codomainY is neither needed nor sufficient.

E — Execution

Carry out the numbered deductions and publish E2/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N2.1 If $s < t$ belong to a connected subset $C \subset R$ but $y \in (s, t)$ does not, then $C \cap (-\infty, y)$ and $C \cap (y, \infty)$ are disjoint nonempty relative open sets coveringC, a contradiction.

N2.2 Thus C contains every point between any two of its points.

N2.3 For a continuous function on a real interval, connectedness of the domain and node1 make its image such a subset; every value between $f(a)$ and $f(b)$ lies in the image.

N2.4 The endpoint values themselves are already attained.

Export E2/v1. A topological proof of the intermediate value theorem.

Acceptance gate. Connectedness of the codomainY is neither needed nor sufficient. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N2: Connectedness of the codomainY is neither needed nor sufficient. Withhold this terminal result until its gate passes.

IV. CONTINUOUS MATHEMATICAL PROOF

Read these steps in order as one proof; the diagram is optional. Numbered activities identify the specific constructions and deductions owned by each node.

A. §4.N1 — Pull back a putative separation

1. Suppose $f(X) = U \cup V$ is a separation in its subspace topology.
2. Write $U = f(X) \cap O$ and $V = f(X) \cap P$ for open $O, P \subset Y$.
3. Their inverse images are open in X by continuity.
4. Both are nonempty because U,V are nonempty subsets of the image.
5. They are disjoint and coverX.
6. This contradicts connectedness.
7. Therefore $f(X)$ is connected; the empty image, if X is empty, also has no separation.

B. §4.N2 — Connected subsets of R are intervals

1. If $s < t$ belong to a connected subset $C \subset R$ but $y \in (s, t)$ does not, then $C \cap (-\infty, y)$ and $C \cap (y, \infty)$ are disjoint nonempty relative open sets coveringC, a contradiction.
2. Thus C contains every point between any two of its points.
3. For a continuous function on a real interval, connectedness of the domain and node1 make its image such a subset; every value between $f(a)$ and $f(b)$ lies in the image.
4. The endpoint values themselves are already attained.

□

V. WORKED ORCHESTRATION

The continuous map $f(t) = (cost, sint)\ from[0, 2\pi]$ has image S^1, so the circle is connected. For $f(t) = t^2\ on[-1, 2]$, the image[0,4] contains every value between the selected endpoint values 1 and 4, even though the map is not injective.

A. Boundary and hypothesis audit

A map from the disconnected two-point discrete space to R with values -1 and 1 is continuous but misses0. Validation identifies the domain condition rather than assuming the codomain's connectedness proves connectedness of every image.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Failure and correction

This is an authored teaching trace, not a recorded agent run.

1. **Contract → Knowledge:** register the §1 theorem, imports, and requested exports before beginning the proof.
2. **Strategy → Execution:** choose the bridges in §3 and carry out the numbered activities in §4; transmit only established E-exports along the §2 DAG.
3. **Validation → responsible node:** test the specific boundary/failure in §5 and the node's individual audit. A failing candidate is withheld; downstream nodes may not use it.
4. **Knowledge + Strategy:** identify whether the error is an omitted condition, an invalid inference, or an attempted extension beyond the theorem. Restore the exact contract or explicitly state a different one.
5. **Execution:** redo the affected mathematical activity using the repaired condition or inference. Recheck consuming nodes, rather than feeding their conclusions back as premises.
6. **Contract:** publish only the supported export with its assumptions and proof location. A hypothetical revised export is v2; the written proof here is the v1 exposition, not a claim that this repair was executed.

B. Hypothesis boundary

A map from the disconnected two-point discrete space to R with values -1 and 1 is continuous but misses0. Validation identifies the domain condition rather than assuming the codomain's connectedness proves connectedness of every image.

C. Version and recheck

Treat any changed assumption as a new contract. Recheck the revised export and all of its descendants. A backward diagnostic never establishes an earlier theorem. This is an authored teaching trace, not an execution log.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading.

- MIT Introduction to Topology: separation, Urysohn, and Tychonoff material.